# Notes on rCCA Reproduction Results

When reviewing the attached notebook (`rcca_combinedwith_s2s_firstresults.ipynb`) and comparing my output to the results from the original paper, you will notice that my results are scientifically consistent, though the exact numerical skill scores differ slightly.

Here is a visual comparison of the extracted cluster centers:

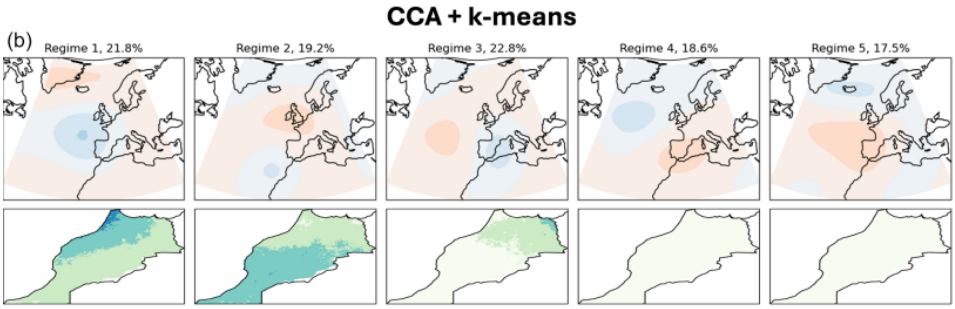


Figure 1: Cluster centers from the original paper.

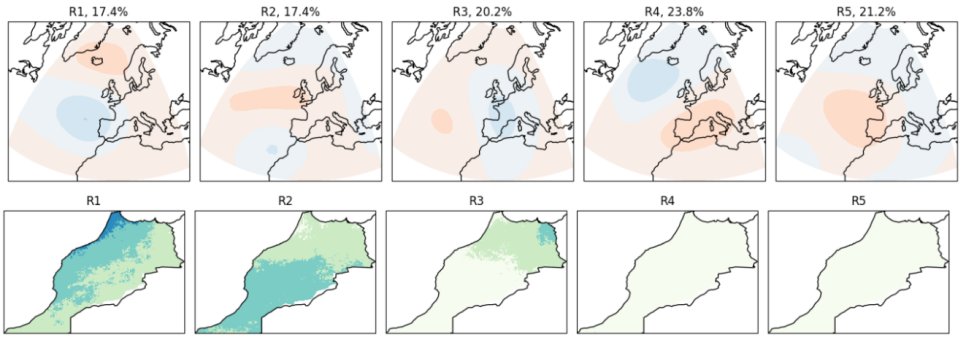


Figure 2: My reproduced cluster centers.

Below is an explanation of the challenge I encountered during the reproduction process and how I resolved it:

## 1. The Missing Data Problem

The main issue I faced was that the original CCA + K-Means pipeline relied on a specific dataset that was missing from our repository: `total_pr_admin_areas.nc`.

According to Section 2.2.2 of the paper:

> *We therefore aggregated precipitation at the district level over Morocco and implemented a ridge regularization parameter for the geopotential height field.*

Because this pre-processed file containing precipitation data explicitly averaged over exact Moroccan administrative districts was not available, I could not feed the exact same "second view" into the Regularized Canonical Correlation Analysis (rCCA).

## 2. My Workaround Solution

To allow the methodology to run and produce meaningful results without requiring new dependencies or external shapefiles, I implemented a **spatial chunking approximation**.

Instead of using the missing administrative regions, my code derives 10 arbitrary, contiguous spatial chunks directly from the available `CHIRPS_pr_total_EW.nc` precipitation field. I flattened the spatial dimension, divided it into 10 equal segments, and took the mean of each chunk.

I then applied the identical Box-Cox transformation to these 10 chunks and fed them into the rCCA algorithm.

## 3. Note on S2S Forecast Evaluation

The final cell of the notebook, which applies the derived clusters to the Subseasonal-to-Seasonal (S2S) hindcast models, could not be executed. This step requires the historical forecast GRIB datasets (`plev_data_perturbed_members_*.grb` and `plev_data_unperturbed_control_member_*.grb`), which are exceptionally large and were not available locally in the `../data.nosync/s2s_hindcasts_LR_atlantic/` directory.